INTEGRATED TOMATO DROUGHT PHENOTYPING

From Canopy Heat to Stomatal Form

A combined SeedThermal and StomatalPhenotypingPipeline research report

16 tomato varieties | 32 plants | 2 water treatments

Prepared for Sapkota Lab

29 July 2026

SCOPE An integrated, quality-aware interpretation of thermal canopy traits, projected plant area, and manually measured stomatal-complex area from the same tomato cohort.

DECISION BRIEF

Executive summary

32	16	85	14
PLANT UNITS	VARIETIES	MANUAL MASKS	THERMAL QC PAIRS

PRIMARY INTERPRETATION The current batch does not show a clear overall treatment effect on canopy temperature or stomatal-complex area. The strongest signal is a provisional increase in projected plant area under drought among thermal-mask-QC-valid pairs.

What the data show

Canopy temperature. Across all 16 pairs, drought-stressed plants averaged 25.69 °C versus 25.84 °C for watered plants (difference -0.15 °C; p = 0.448). Two plant masks failed QC, so this full-set estimate is descriptive.

Projected plant area. After restricting to 14 complete QC-valid pairs, drought-stressed plants had 39.3% larger projected area on average (difference 2,374 pixels²; 95% CI 968 to 3,779; p = 0.003). This remains provisional pending camera-geometry review and physical calibration.

Stomatal-complex area. Drought-stressed plants averaged 10,211.5 pixels² versus 10,968.5 pixels² for watered plants. The paired difference was -757.1 pixels² (-6.9%; 95% CI -2,766.3 to 1,252.2; p = 0.434).

Cross-scale relationship. Plant-level stomatal area was essentially unrelated to canopy temperature or projected plant area in this batch. The two pipelines are complementary rather than interchangeable.

Recommended use

Use now	Descriptive treatment and variety profiles; candidate prioritization; protocol refinement.
Do not claim yet	Generalized drought-response mechanisms, calibrated physical area, stomatal density, or variety ranking with inferential certainty.
Next milestone	Correct two thermal masks, acquire physical scale references, and repeat with biological replication.

ONE COHORT, TWO SCALES

Study design and evidence chain

SeedThermal and StomatalPhenotypingPipeline are treated as a single experiment. Every row resolves to a stable plant unit, a variety, and one of two biological treatment labels.

Watered	Source code Regular; 100 mL.
Drought-stressed	Source code Control; 0 mL.
Experimental unit	One plant. Individual stomata are within-plant subsamples, not independent replicates.
Pairing	One watered and one drought-stressed plant for each of 16 varieties.

Evidence collected

Scale	Input	Primary traits	Usable scope
Canopy	64 radiometric FLIR captures across 13-15 July	Mean, median, range, dispersion, ΔT-air	July 15 isolates 32 plants; earlier paired frames show combined canopies.
Plant silhouette	32 July 15 binary masks	Projected area and frame fraction	30 masks pass QC; 14 complete variety pairs.
Leaf epidermis	32 micrographs	Candidate confidence and per-object evidence	5,633 candidates retained across decision tiers.
Stomatal complex	85 manual polygons	Area, length, width, perimeter, shape, orientation	2-3 quality-selected objects per plant; pixel units.

Source: combined plant manifest, thermal run table, calibrated candidate ledger, and manual mask manifest.

Thermal acquisition context

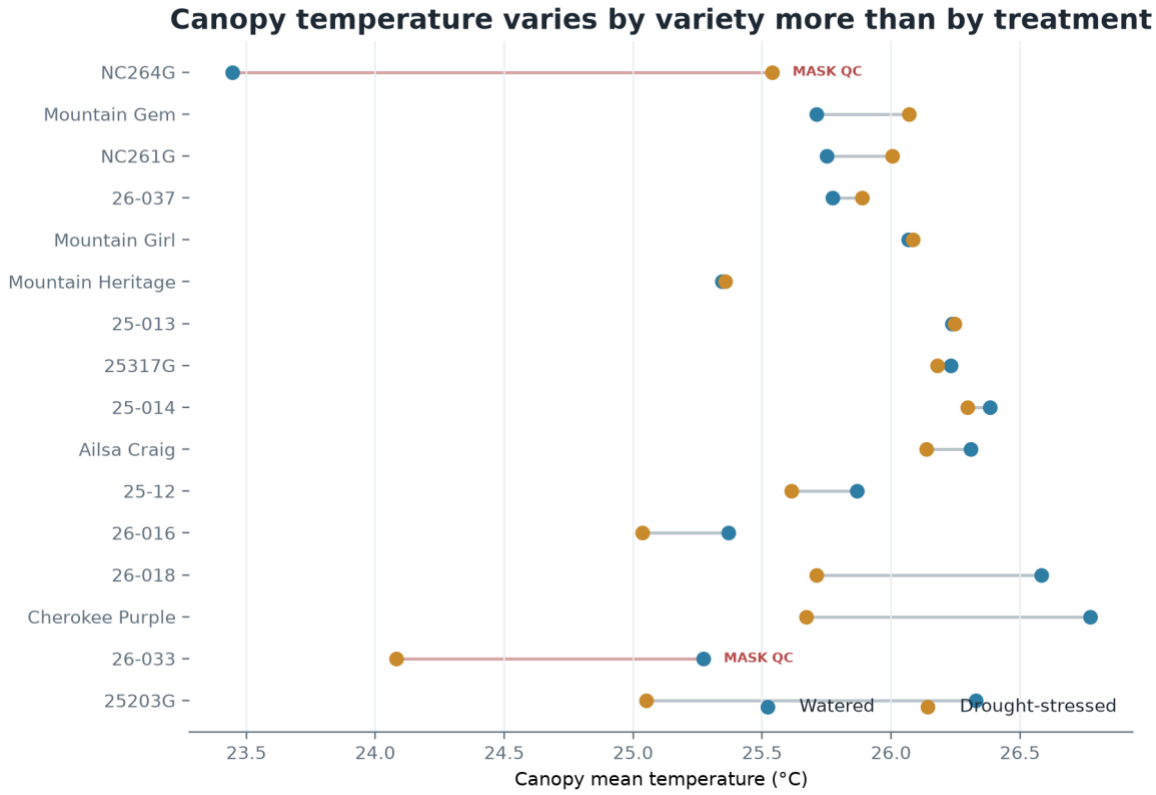
Date	Captures	Mean °C	Range °C	Interpretation
2026-07-13	16	24.32	23.86-24.56	Paired full-frame: both treatments in one image
2026-07-14	16	23.57	23.23-24.08	Paired full-frame: both treatments in one image
2026-07-15	32	25.76	23.45-26.77	Individual auto-masked plants; treatment comparison

CANOPY PHENOTYPE

Thermal findings

BOTTOM LINE No clear overall treatment-temperature difference was detected. Across all 16 pairs, the mean drought-minus-watered difference was -0.15 °C (95% CI -0.58 to +0.27; p = 0.448).

Figure 1. Paired canopy temperatures by variety. Red QC labels mark the two pairs affected by failed automatic masks.



Source: July 15 plant-level FLIR extraction. Watered = Regular; drought-stressed = Control.

QC-restricted sensitivity result

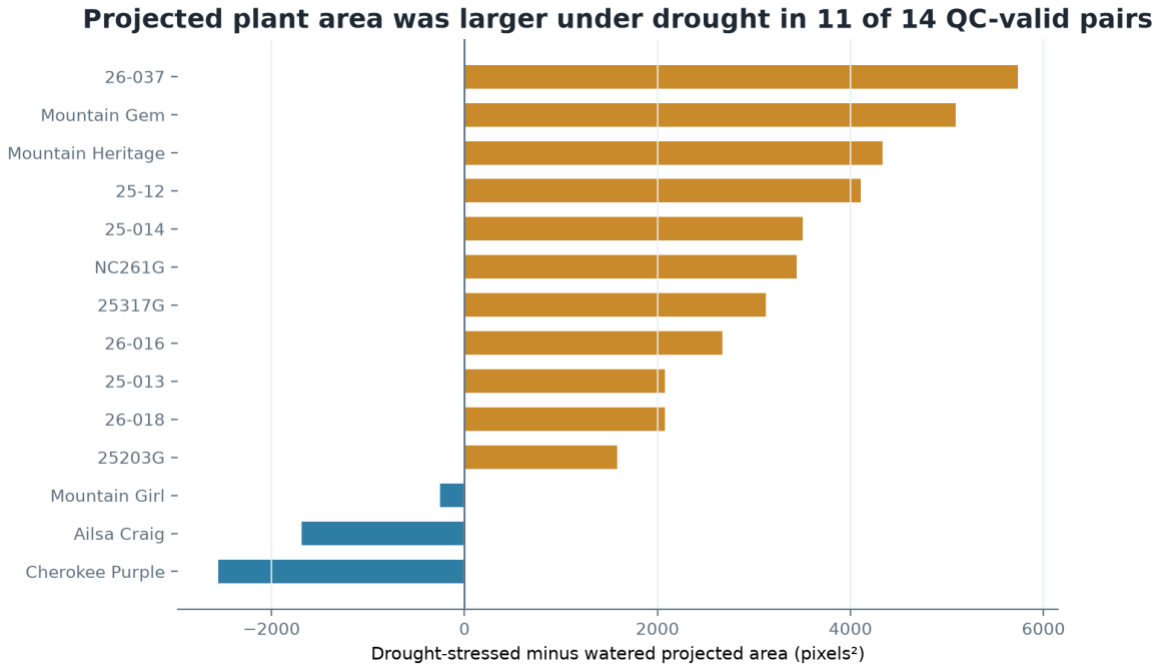
After excluding NC264G and 26-033 to retain 14 complete QC-valid pairs, drought-stressed plants averaged 25.81 °C and watered plants 26.05 °C. The paired difference was -0.24 °C (95% CI -0.53 to +0.05; paired t-test p = 0.093; Wilcoxon p = 0.217). This near-zero-to-modest cooling estimate should be treated as inconclusive rather than as evidence of equivalence.

PLANT SILHOUETTE PHENOTYPE

Projected plant area

6,035	8,409	+39.3%	0.003
WATERED MEAN PX²	DROUGHT MEAN PX²	PAIRED DIFFERENCE	PAIRED T-TEST P

Figure 2. Within-variety difference in projected plant-mask area among the 14 complete thermal-QC pairs.



Blue bars favor watered; ochre bars favor drought-stressed. Area is in image pixels, not physical units.

Drought-stressed plants were larger in 11 of 14 QC-valid pairs. The mean paired difference was 2,374 pixels² (95% CI 968 to 3,779); paired t-test p = 0.003, Wilcoxon p = 0.005.

WHY THIS REMAINS PROVISIONAL The signal is statistically coherent within this batch, but projected pixel area depends on mask quality, camera distance, angle, and plant pose. It is not yet calibrated leaf area.

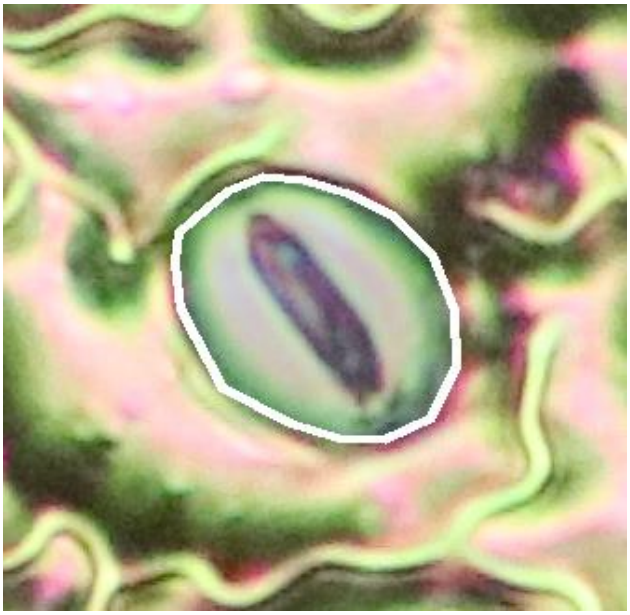
OBJECT-LEVEL EVIDENCE

Stomatal detection and measurement

5,633	325	1,495	3,813
CANDIDATES	ACCEPTED TIER	REVIEW TIER	REJECTED TIER

The automated system is structured as an analysis engine: it preserves every candidate, its model evidence, confidence-related fields, decision tier, image position, and measurement context. Candidate counts are a derived summary, not the primary scientific output.

Figure 3. Example of a quality-selected stomatal complex with the manually traced outer contour.



White boundary = reviewer polygon; translucent teal = measured stomatal-complex region.

Manual measurement subset

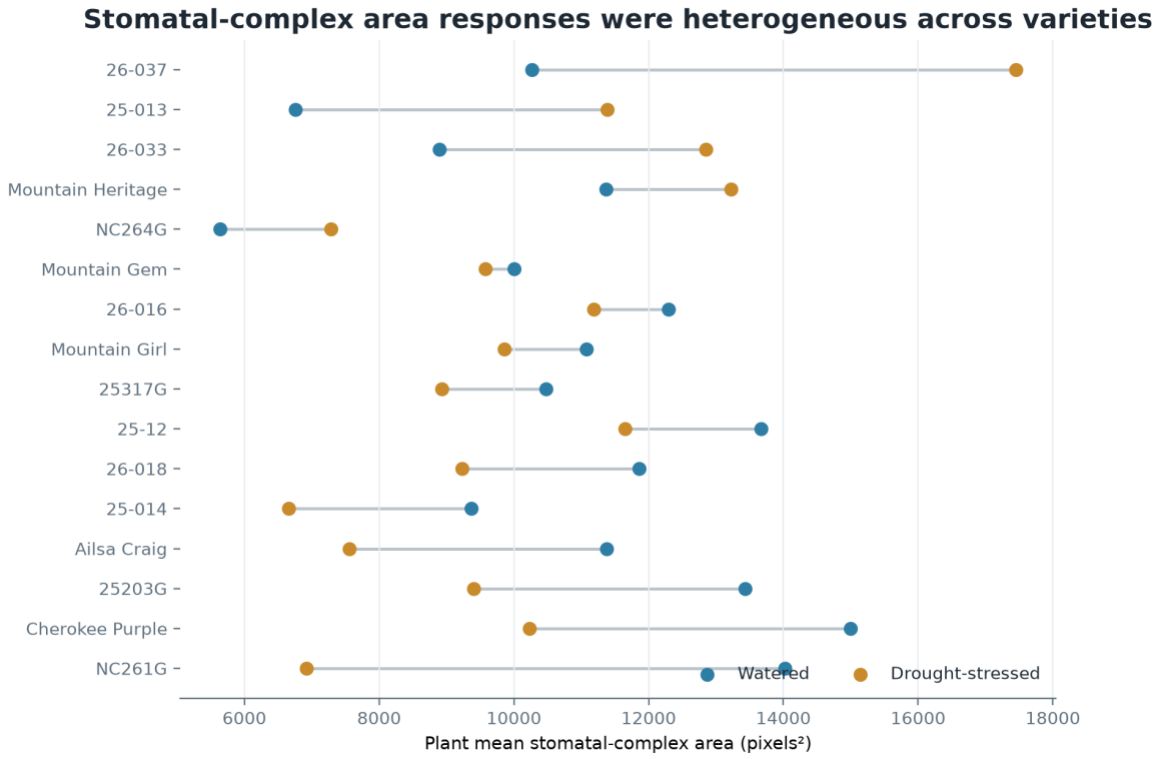
Coverage	85 accepted manual polygons across 32 plants and all 16 varieties; 2-3 objects per plant.
Selection	Highest local focus and contrast with low clipping and good edge containment among completed masks.
Not used for selection	Treatment, variety, object area, mask area, or apparent stomatal shape.
Available traits	Complex area, length, width, perimeter, aspect ratio, eccentricity, solidity, circularity, orientation, centroid.
Current unit	Pixels and pixels^2. Physical dimensions require a stage-micrometer calibration.

MICROSCOPIC PHENOTYPE

Stomatal-complex area by treatment

10,968.5	10,211.5	-6.9%	0.434
WATERED PX²	DROUGHT PX²	RELATIVE DIFFERENCE	PAIRED T-TEST P

Figure 4. Plant-level mean stomatal-complex area for each of 16 matched variety pairs.



Each plant mean is based on 2-3 quality-selected manual masks; inferential tests use plants, not individual stomata.

The mean paired difference, defined as drought-stressed minus watered, was -757.1 pixels². Its 95% confidence interval ranged from -2,766.3 to 1,252.2 pixels². The paired t-test gave p = 0.434; the Wilcoxon signed-rank test gave p = 0.404; Cohen's dz = -0.20. The estimate is compatible with both modest drought-associated reduction and modest increase.

INTERPRETATION There is no clear batch-level treatment effect on stomatal-complex area. The variety-specific directions are heterogeneous and should guide replication, not selection decisions.

GENOTYPE PROFILES

Descriptive comparison among 16 varieties

The highest two-treatment descriptive mean was observed for 26-037; the lowest for NC264G. Because each variety-treatment cell contains one plant, these are profiles of this cohort, not population estimates.

Variety	Watered px²	Drought px²	Two-treatment mean	Drought - watered
26-037	10,269	17,451	13,860	+7,183
25-12	13,670	11,652	12,661	-2,018
Cherokee Purple	14,993	10,230	12,611	-4,762
Mountain Heritage	11,369	13,223	12,296	+1,854
26-016	12,299	11,188	11,744	-1,111
25203G	13,436	9,401	11,418	-4,035
26-033	8,892	12,849	10,870	+3,957
26-018	11,857	9,230	10,543	-2,627
NC261G	14,028	6,918	10,473	-7,110
Mountain Girl	11,072	9,860	10,466	-1,212
Mountain Gem	10,005	9,574	9,789	-431
25317G	10,477	8,928	9,702	-1,550
Ailsa Craig	11,374	7,561	9,467	-3,812
25-013	6,753	11,383	9,068	+4,630
25-014	9,363	6,656	8,010	-2,708
NC264G	5,641	7,280	6,461	+1,639

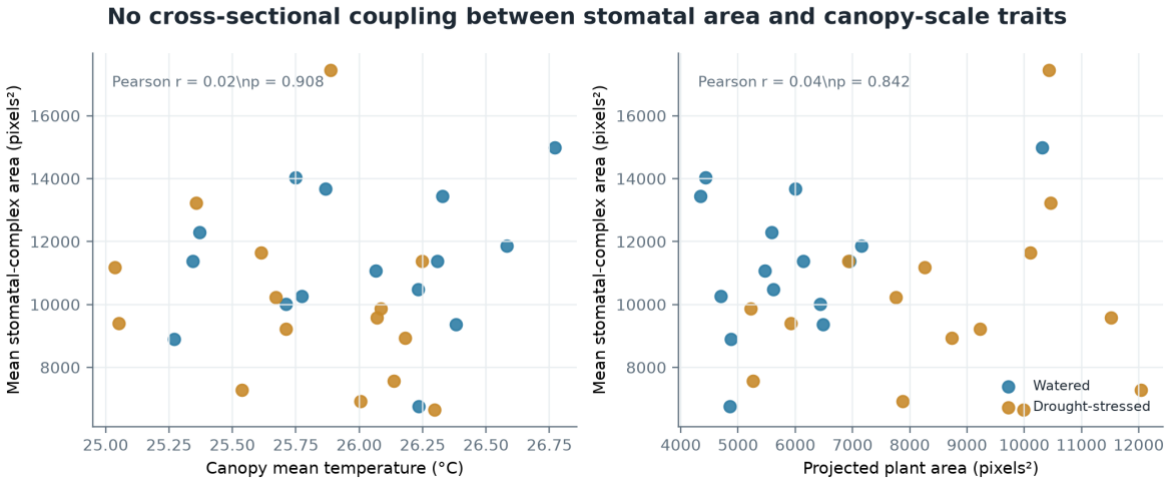
Ranking is based on the descriptive average of the watered and drought-stressed plant means.

RESPONSE RANGE The largest positive response was 26-037 (+7,183 pixels²); the largest negative response was NC261G (-7,110 pixels²). These extremes are especially sensitive to low replication.

CROSS-SCALE SYNTHESIS

Do canopy and stomatal traits move together?

Figure 5. Plant-level stomatal-complex area versus thermal and projected-area traits for the 30 thermal-QC-valid plants.



Correlations are exploratory; points are plants and colors indicate treatment.

Stomatal area vs canopy temperature	Pearson r = 0.022; p = 0.908; n = 30 plants.
Stomatal area vs projected plant area	Pearson r = 0.038; p = 0.842; n = 30 plants.
Paired stomatal response vs thermal response	Pearson r = 0.379; p = 0.181; n = 14 varieties.
Paired stomatal response vs projected-area response	Pearson r = 0.471; p = 0.089; n = 14 varieties.

The cross-sectional relationships are essentially null. The response-level associations are moderate in direction but imprecise, especially for projected area (r = 0.47, p = 0.089). They are useful hypotheses for a replicated experiment, not evidence of mechanistic coupling.

SCIENTIFIC READING A plant can show canopy-scale size or temperature differences without a matching change in the area of its clearest stomatal complexes. Response is multiscale; one trait cannot substitute for another.

INTERPRETATION GUARDRAILS

Limitations and recommended next steps

Constraint	Why it matters	Recommended action
One plant per variety-treatment cell	Variety effects and interactions cannot be estimated reliably.	Repeat with independent biological replicates.
2-3 selected stomata per plant	Within-leaf variation is only sparsely represented.	Expand objects per plant using the validated analysis engine.
Quality-selected objects	Results describe clearest measurable stomata, not a random within-image population.	Predefine random/systematic fields and selection rules.
Pixel-only geometry	Areas cannot be compared in µm² or across changed magnification.	Acquire a stage-micrometer image at every optical configuration.
Two failed thermal masks	NC264G watered and 26-033 drought distort area and temperature extraction.	Correct masks and rerun the joined analysis.
Projected area is view-dependent	Camera pose and plant architecture affect silhouette size.	Fix camera geometry and add a scale reference.
No stomatal density or pore aperture	Current masks support complex morphology, not density or conductance.	Define analyzed leaf area; validate pore segmentation separately.

These constraints define the boundary between current evidence and future claims.

Prioritized next experiment

1 Repair	Correct the two thermal masks and regenerate all joined outputs.
2 Calibrate	Capture stage-micrometer and plant-view scale references.
3 Replicate	Target at least 3-5 plants per variety-treatment cell; preserve within-variety pairing.
4 Sample	Use predefined fields and a consistent number of stomata per plant.
5 Validate	Reserve held-out tomato images for detection calibration and mask-error assessment.
6 Extend	Only after validation, add pore aperture, density, and spatial-pattern traits.

GO/NO-GO RECOMMENDATION Proceed with the combined platform as a research workflow. Do not yet use this single batch to rank varieties for drought tolerance or claim a stomatal mechanism.

REPRODUCIBILITY

Methods and data provenance

Thermal pipeline

Radiometric FLIR JPGs were converted to per-pixel Celsius arrays. July 15 plant masks defined the canopy pixels used for mean, median, minimum, maximum, standard deviation, P10, P90, and temperature relative to the 25 °C ambient reference. Projected plant area is the number and fraction of selected pixels.

Stomatal pipeline

Native-resolution micrographs were processed with a high-recall, tiled candidate detector. The object ledger retains decision tiers and confidence evidence. For this report, manual outer stomatal-complex polygons were ranked within each image by local focus, contrast, clipping, and edge containment; at most three objects per plant were retained. Polygon geometry produced direct pixel-space morphology.

Statistical analysis

Treatment contrasts were paired by variety. Plant means, not individual stomata, were the inferential units. Paired t-tests were accompanied by 95% confidence intervals and Wilcoxon signed-rank sensitivity tests. Variety results are descriptive because each variety-treatment cell contains one plant. Cross-modal Pearson correlations were exploratory and restricted to thermal-mask-QC-valid plants or pairs.

Primary source files

Combined plant table	results/combined_quality_mask_analysis_v1/plant_phenotypes.csv
Variety-level paired table	results/combined_quality_mask_analysis_v1/variety_phenotypes.csv
Per-stoma measurements	results/quality_mask_analysis_v1/stomatal_complex_measurements.csv
Stomatal statistical record	results/quality_mask_analysis_v1/area_comparisons/stomatal_area_analysis.json
Thermal run table	../SeedThermal/processed/thermal_data_all_sessions.csv
Study definitions	docs/COMBINED_STUDY.md and docs/ANALYSIS_ENGINE.md

Reporting language

PREFERRED CONCLUSION In this quality-selected pilot cohort, drought stress was not associated with a clear overall change in canopy temperature or stomatal-complex area. A larger projected plant silhouette was observed in the QC-valid drought group, but this finding remains provisional pending mask correction, calibration, and biological replication.